UCLA Math 131A, Winter 2025
Midterm 2, Lecture 3

Instructor: Linfeng Li

Friday, February 21

Please print your name and student ID below.

Name: ______________________________

Student ID: ___________________________

Please read all of the following rules before proceeding.

- The exam worth 40 points in total.
- Show all your work, and precisely state any theorems that you use.
- You are not allowed to use calculators or notes during the test, but you may use one-page, double-sided, and handwritten cheat sheet.
- Good luck!

Question	Points	Score
1	10	
2	5	
3	9	
4	8	
5	8	
Total:	40	

1. (10 points) Circle TRUE or FALSE for each of the following statement.

   (i) $\boxed{\text{True}}$ False A sequence in $\mathbb{R}$ is Cauchy if and only if it is convergent.

   (ii) True $\boxed{\text{False}}$ Every bounded sequence is convergent.

   (iii) $\boxed{\text{True}}$ False Let $(s_n)_{n\in\mathbb{N}}$ be a sequence with $s_n \in [c, d], \forall n \in \mathbb{N}$, where $c < d$. Then $\limsup s_n \in [c, d]$.

   (iv) $\boxed{\text{True}}$ False Every subsequence of an increasing sequence is increasing.

   (v) True $\boxed{\text{False}}$ A sequence $(s_n)_{n\in\mathbb{N}}$ converges if and only if $(s_n^2)_{n\in\mathbb{N}}$ converges.

2. (5 points) In this problem, we aim to prove that

$$\lim_{n\to\infty} \frac{\sqrt{n} - 2025}{\sqrt{n} - 2024} = 1$$

by definition. Let $\epsilon > 0$. Find an $N \in \mathbb{N}$ such that

$$\left| \frac{\sqrt{n} - 2025}{\sqrt{n} - 2024} - 1 \right| < \epsilon$$

for all $n \geq N$.

Solution: For $\epsilon > 0$, let $N \in \mathbb{N}$ be such that $N > (1/\epsilon + 2024)^2$. Then for $n \geq N$, we have

$$\sqrt{n} - 2024 \geq \sqrt{N} - 2024 > \frac{1}{\epsilon} > 0$$

and

$$\left| \frac{\sqrt{n} - 2025}{\sqrt{n} - 2024} - 1 \right| = \frac{1}{|\sqrt{n} - 2024|} = \frac{1}{\sqrt{n} - 2024} \leq \frac{1}{\sqrt{N} - 2024}$$
$$< \frac{1}{1/\epsilon + 2024 - 2024} = \epsilon.$$

3. Let $(s_n)_{n\in\mathbb{N}}$ be a sequence defined by $s_1 = 10$ and

$$s_{n+1} = \frac{1}{4}(s_n + 6), \quad \forall n \in \mathbb{N}.$$

(a) (3 points) Prove that $s_n > 2$ for all $n \in \mathbb{N}$.

(b) (3 points) Prove that $(s_n)_{n\in\mathbb{N}}$ is a decreasing sequence.

(c) (3 points) Determine convergence or divergence of the sequence $(s_n)_{n\in\mathbb{N}}$. If it converges, find the limit.

Solution: (a) Let P_n be the statement that $s_n > 2$. The base case P_1 is obvious since $s_1 = 10 > 2$. We assume P_n holds for some $n \in \mathbb{N}$. We compute

$$s_{n+1} = \frac{1}{4}(s_n + 6) > \frac{1}{4}(2 + 6) = 2.$$

Hence, P_{n+1} holds. By induction, $s_n > 2$ for all $n \in \mathbb{N}$.

(b) From part (a) we get, for any $n \in \mathbb{N}$,

$$2 \le s_n \implies s_n + 6 \le 4s_n \implies \frac{1}{4}(s_n + 6) \le s_n \implies s_{n+1} \le s_n.$$

Hence, the sequence (s_n) is decreasing.

(c) The sequence (s_n) is decreasing and bounded from below, by MCT, the sequence converges. Let $s = \lim_{n\to\infty} s_n$. From the iterative relation and limit law, we get

$$\lim_{n\to\infty} s_{n+1} = \frac{1}{4}\lim_{n\to\infty}(s_n + 6) \implies s = \frac{1}{4}(s + 6) \implies s = 2.$$

4. Let $(s_n)_{n\in\mathbb{N}}$ be a sequence defined by

$$s_n = (-1)^n + \frac{1}{n}, \quad \forall n \in \mathbb{N}.$$

(a) (4 points) Show that $(s_n)_{n\in\mathbb{N}}$ diverges. You may use any method.

(b) (4 points) Construct two subsequences that converge to two different limits.

Solution: (a) For the sake of contradiction, we assume (s_n) converges. Since the sequence $(\frac{1}{n})$ converges, by limit law, the sequence $(s_n - \frac{1}{n})$ converges. However, the sequence $(s_n - \frac{1}{n}) = ((-1)^n)$ which diverges. Contradiction!

(b) Consider the subsequences $(s_{2k})_{k\in\mathbb{N}} = (1 + \frac{1}{2k})_{k\in\mathbb{N}}$ and $(s_{2k+1})_{k\in\mathbb{N}} = (-1 + \frac{1}{2k+1})_{k\in\mathbb{N}}$. From class we know the sequence $(1/2k)$ and $(1/(2k+1))$ are convergent and converge to 0. Hence,

$$(s_{2k})_{k\in\mathbb{N}} \to 1$$

and

$$(s_{2k+1})_{k\in\mathbb{N}} \to -1.$$

5. In each of the following situation, decide whether the sequence is convergent. If yes, justify it; otherwise, give an example which diverges.

   (a) (4 points) The sequence $(t_n)_{n\in\mathbb{N}}$ satisfies for all $\epsilon > 0$, there exists some $N \in \mathbb{N}$ such that $|t_N| < \epsilon$. *Note that this is different from the definition of convergence.*

   (b) (4 points) The sequence $(s_n)_{n\in\mathbb{N}}$ satisfies $|s_n - s_m| \leq \frac{1}{n} + \frac{1}{m}$ for all $n, m \in \mathbb{N}$.

Solution: (a) The sequence can be divergent. We consider the sequence $(t_n)_{n\in\mathbb{N}} = (n-1)_{n\in\mathbb{N}} = \{0, 1, 2, \ldots\}$. Then for any $\epsilon > 0$, there exists $N = 1$ such that $|t_N| = |t_1| = 0 < \epsilon$. However, the sequence $(t_n)_{n\in\mathbb{N}}$ diverges to infinity.

(b) For $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $N > 2/\epsilon$. Then for any $n, m \geq N$, we have

$$|s_n - s_m| \leq \frac{1}{N} + \frac{1}{N} = \frac{2}{N} < \epsilon.$$

Hence, the sequence (s_n) is Cauchy by definition and thus convergent.